

Global Standards for Federated Health Data Science: The GA4GH Approach

When the data cannot travel, send the analysis instead

Team 3: Rosetta

1 YOUR FLATMATE

Your flatmate is rushed to A&E with severe stomach pain. She is not responding to any antibiotic she has been given. So, to understand what is growing in her gut, a sample of her gut bacteria is sequenced. The sequence has to be compared against a far larger reference collection, held by a research group in another country. It cannot come to her. **It is too large to move, governed by different data-sharing laws, and stored in a different format.**

Building a one-off system to bridge the two could take months. She does not have months. **This is an example of the type of problem GA4GH exist to solve. To speed up the progress in genomic research and human health.**

2 WHY THIS KEEPS HAPPENING

How many persons bought a phone because of the charging port? Exactly. Rival technologies and the EU Commission had the same thought and they made USB-C the standard.

This is pre-competitive standard. Health research had no equivalent, so every collaboration was built from scratch which limits the progress of research and human health care.

3 WHO AGREES THE STANDARDS

The **Global Alliance for Genomics and Health (GA4GH)** is an international community that develops shared standards, policy frameworks and tools for genomic and health data.

It holds no data itself. Hospitals and research groups keep their own, and remain responsible for it. GA4GH's job is to agree the common rules so nobody has to invent them twice.

4 THE STANDARD WE STUDIED

You want to bake a cake. But the ingredients are in Australia. So, you send the recipe to Australia. The cake is baked there and is then sent to you in the UK

GA4GH's Task Execution Service (TES) is the recipe: one agreed structure for describing a computational task, what data it needs, what software runs and what comes back, so any compute system can read it.

TES is only one standard, alongside WES (whole workflows), Beacon (finding data) and Passports (access).

5 WHO IS PUTTING IT TO WORK

ELIXIR is Europe's life-science research infrastructure, organised as national Nodes. It has worked with GA4GH since 2013, formalised as a Strategic Partnership in 2019.

ELIXIR Cloud & AAI is a GA4GH **Driver Project**, a real initiative that helps test and implement GA4GH standards. It is building a federated compute network across those Nodes, with TES at its core.

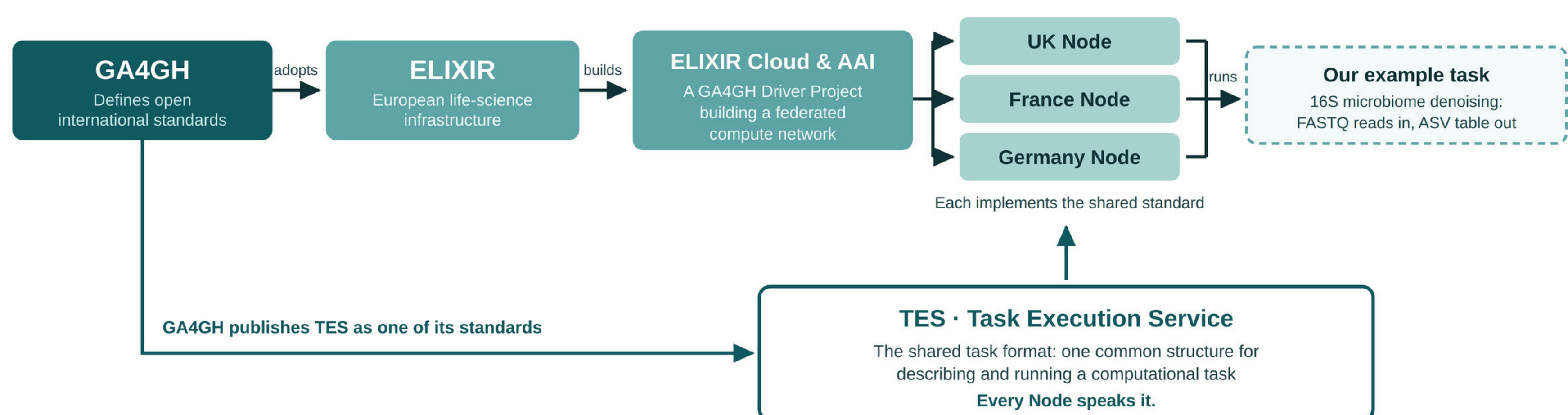
6 BACK TO YOUR FLATMATE

With a shared standard in place, nobody builds anything from scratch. Her lab describes the analysis **once**, as a TES task document.

It travels to the Node holding the reference collection, runs beside the data, and only the result comes back. **The data never moves.**

Illustrative scenario. Sequencing identifies which bacteria are present; treatment decisions rest on wider clinical testing.

7 HOW IT FITS TOGETHER



8 WHAT OUR TEAM BUILT

- An example TES task document, a three-step microbiome analysis, validated against the official GA4GH schema with no violations
- A plain-language explainer, a ten-question FAQ and a 35-term glossary
- A source log tying 25 claims to the official page each came from

! AN IMPORTANT LIMIT

Validation proves the document's **shape** is right, not its **science**. A typo in a filename would pass every check and fail only when the task ran.

Our example uses synthetic data and has not been run on a live service.

Everything here is openly published. The GA4GH standards, the TES specification and the schema we validated against are all free to read.

Sources: GA4GH, About us · GA4GH Driver Projects, ELIXIR Cloud and AAI · GA4GH, ELIXIR and GA4GH expand collaboration, 2 May 2019 ·

GA4GH, Task Execution Service · github.com/ga4gh/task-execution-schemas. Accessed 19 August 2026.

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team3_TES_Example